Curtailment traps early green hydrogen assets beyond the reach of learning

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Summary

Green hydrogen is expected to allow early electrolysis plants to inherit industry-wide cost declines through replaceable stacks. We test this channel using 10,214 operating Chinese wind and utility-scale solar records with reconstructed hourly low-opportunity-cost electricity, six ERA5 weather-year replays and 30-year nominal equity cash flows. Two obstacles compound. Curtailment lowers electricity cost but concentrates operation, so 701 of 710 marginal records never reach the 60,000-h replacement threshold. Fundamentally, even where replacement occurs, the replaceable stack embodies only 18.2% (range 11.3–42.3%) of the capital saving achieved by a 2060 new build, and across a 1–10% return frontier sector learning repairs marginal returns only where the initial shortfall is small. Price convergence deepens the loss but is not required. Replaceability is therefore neither sufficient nor decisive, and durable de-risking depends less on deployment volume than on verifiable power-supply commitment and pre-investment capacity flexibility before capital becomes irreversible.

Context & scale

Electrolyser costs are projected to fall as deployment expands, and replaceable stacks are widely assumed to pass these gains to plants already in operation. This assumption is consequential for green hydrogen projects built around curtailed renewable power: cheap electricity can make them appear investable, yet sparse operating hours may prevent the first stack replacement. Using 10,214 Chinese wind and utility-scale solar records and six weather-year replays, we show that 701 of 710 marginal records never access stack-mediated learning and that the replaceable stack carries only 18.2% of 2060 new-build capital savings in the central case. Deployment support alone therefore cannot repair weak incumbent returns. De-risking should occur before investment by verifying power-supply commitments and preserving capacity flexibility; transport, demand and water constraints remain project-specific limits.

Highlights

- Curtailment prevents 701 of 710 marginal records from replacing their stacks
- Replaceable stacks embody 18.2% of 2060 new-build capital savings
- Sector learning repairs returns only when the initial shortfall is small

- Power commitments and capacity flexibility move risk before investment

Keywords: green hydrogen; low-opportunity-cost electricity; project finance; learning curves; vintage capital; investment flexibility

Introduction

Green hydrogen is central to deep decarbonisation in steel, chemicals, refining and long-distance transport[1–3]. The International Energy Agency's net-zero pathway requires low-emissions hydrogen production to rise from about 1 Mt today to roughly 70 Mt by 2030, while the 2024 project pipeline contained nearly 520 GW of announced electrolysers but only 4% had reached final investment decision or construction[4, 5]. IRENA estimates that green-hydrogen production and infrastructure require cumulative investment of about US$3.8 trillion by 2050[6]. China accounted for more than 40% of electrolysis capacity reaching final investment decision during the preceding year and 60% of manufacturing capacity[7–10].

Embodied technical change provides the natural starting point. In vintage-capital theory, improvements embodied in new equipment do not automatically raise sunk-capital productivity[11]. Experience curves link cumulative deployment to declining unit cost[12–14]. Empirical work shows that aggregate learning rates can conceal component-specific drivers[15–17], although broad deployment–cost regularities persist across technologies[18, 19]. Electrolysis studies project substantial sectoral cost and performance gains[20–23], while technical assessments divide those gains among stacks, balance of plant and operating performance[24–26]. Green-hydrogen deployment models then apply these sector-level trajectories to future capacity[27–29]. Electrolysis is an unusually permissive test because its stack is a large, replaceable component. Later cost, efficiency and lifetime gains can therefore enter an incumbent plant at replacement. Asset appraisal nevertheless needs vintage-specific incidence. The unresolved question is how much sector decline an already commissioned asset can physically capture.

Curtailment creates a conflict between entry cost and access to learning[30–32]. Low-opportunity-cost electricity can make electrolysis appear attractive at commitment, yet its temporal scarcity suppresses annual operating hours. Those hours determine whether a stack reaches its replacement threshold within the project life and hence whether later cost, efficiency and lifetime improvements enter the cash flow. The same resource condition that lowers the initial electricity bill can sever the replacement-mediated route to subsequent learning. We term this the *curtailment-learning trap*: sector learning succeeds, but a sparsely operated incumbent remains locked to its commissioning vintage. Price convergence can deepen the loss, but it is not required for the mechanism.

We test this mechanism by coupling 10,214 operating wind and utility-scale photovoltaic records with reconstructed hourly low-opportunity-cost electricity, electrolyser sizing, 30-year nominal equity cash flow and vintage-specific learning; six ERA5 years provide a meteorological replay. Two disclosed firm rules define a lower screening case and a 6.5% comparison within a continuous 1–10% return frontier[33–35]. In the reference reconstruction, 710 records qualify only under the lower rule. Of these, 701 fail to reach the 60,000-h stack-replacement threshold, central operating learning raises only three above 6.5% at constant hydrogen price, and favourable transfer of all contemporaneous non-stack savings raises the count to eight. The central decomposition assigns 18.2% of a 2060 new-build capital saving to the replaceable stack, with a source-grounded boundary of 11.3–42.3%. Utilisation, not deployment scale alone, is the gate through which later learning reaches an early asset.

Results

Low-opportunity-cost electricity, rather than total renewable output, limits developable supply

System-constrained electricity compresses the physical hydrogen boundary within the covered inventory to about 6% of its full-output upper bound. The inventory contains 4,149 operating wind records and 6,065 operating utility-scale photovoltaic records, totalling approximately 629 GW. It represents 72% of June 2024 official all-wind-plus-centralised-photovoltaic capacity, or 53% when distributed photovoltaics are included. Under the reference resource profile, the inventory produces about 1,109 TWh, whereas the 2025-calibrated system-constrained quantity is 70.8 TWh. At $55\,\mathrm{kWh\,kg^{-1}\,H_2}$, these mutually exclusive branches correspond to 20.2 and $1.29\,\mathrm{Mt\,H_2\,yr^{-1}}$. The latter is therefore an inventory-conditioned physical boundary, not an estimate of national green-hydrogen potential.

Temporal concentration further reduces operable scale, consistent with China's system-specific renewable curtailment[36]. Under the reference daily peak-shaving reconstruction, the median record has positive constrained electricity in approximately 1,300 hours, and the highest-output 10% of those hours contain about 91% of annual constrained energy (Fig. 1b). Replaying six ERA5 weather years gives inventory-level potentials of 1.28–$1.34\,\mathrm{Mt\,yr^{-1}}$ and pairwise Spearman correlations above 0.996 for record-level resource rankings (Fig. 1d,e). Annual peak-shaving and proportional allocation conserve the same annual energy yet produce median active durations of approximately 460 and 5,180 hours. The annual quantity is calibrated, but its record-level and hourly allocation remains only partially identified.

Maximising nominal energy capture can over-size electrolysers. With a 30% minimum-load constraint for alkaline electrolysis, nominal capture targets of 60% and 90% require approximately 31.7 and 64.3 GW of electrolyser capacity and produce 0.75 and $1.08\,\mathrm{Mt\,H_2\,yr^{-1}}$. Extending the target to 100% raises capacity to about 163 GW, but the larger unit cannot remain above minimum load during many constrained hours and realised production declines to $0.78\,\mathrm{Mt\,yr^{-1}}$. The relevant physical quantity is therefore usable production from the hourly profile, rather than annual electricity capture alone (Fig. 1c).

Economic screening retains only part of this physical boundary. In the static 28 $\mathrm{CNY\,kg^{-1}}$ reference, the lower firm rule qualifies 1,809 records over the 30-year primary horizon, corresponding to 8.4 GW of electrolysers and $0.32\,\mathrm{Mt\,H_2\,yr^{-1}}$, or about one quarter of the constrained-electricity potential. Provincial physical potential is concentrated in Inner Mongolia, Xinjiang, Hebei, Gansu and Qinghai, but qualified supply does not scale proportionally with that potential (Fig. 1a). Holding annual constrained energy fixed while changing only its hourly allocation moves qualified hydrogen supply from 0.12 to $0.81\,\mathrm{Mt\,yr^{-1}}$. Annual resource totals bound the opportunity; hourly concentration, minimum load and cash-flow conditions determine how much of it passes an economic screen.

Return criteria create a marginal band whose size depends on information at commitment

Static price and technology expectations expose a continuous return frontier. At a producer-side price of 28 $\mathrm{CNY\,kg^{-1}}$ with no future operating improvement credited at commitment, the lower rule qualifies 1,809 records and independent optimisation at 6.5% qualifies 1,099. The remaining 710 records, 39% of the lower-rule set, are *strict-marginal*: some capacity has non-negative equity net present value (NPV) under the lower rule, but none reaches 6.5%. Across 972 deterministic configurations, the qualified sets remain separated (Fig. 2a). At the IEA's 2021 China utility-solar cost-of-capital estimate (4.9% nominal), exact revaluation qualifies 1,255 records, compared with 1,036 at 8% and 874 at 10%[37]. This is financing context rather than a green-hydrogen WACC;

the cohort flow and full 1–10% curve show that the band is not created by comparing only two thresholds (Fig. 2b,c).

The lower rule expands project counts more than it expands hydrogen supply. Its 1,809 qualified records correspond to 8.4 GW, 60.7 billion CNY of gross installed-system capital and 0.32 Mt H_2 yr^{-1}. Strict-marginal records account for 1.5 GW, 11.0 billion CNY and 0.048 Mt yr^{-1}, or 18%, 18% and 15% of those totals. Xinjiang and Qinghai contain 393 strict-marginal records and approximately 68% of their capital and 67% of their output (Fig. 2e); capacity–yield scaling confirms exposure across many small configurations (Fig. 2f). These are plant-gate exposures. A provincial delivery-cost screen[38] contracts the two feasible sets to 331 and 93 records and completely changes strict-cohort membership (Supplementary Note 10); the spatial pattern therefore identifies production exposure, not delivered competitiveness.

Expected market conditions narrow the qualified set before investment, but do not remove the wedge between return requirements. If a linear producer-price path to 22 CNY kg^{-1} and accessible operating learning are both known at commitment, 1,282 records qualify under the lower rule and 971 at 6.5%, leaving 311 strict-marginal. With a linear path to 18 CNY kg^{-1}, the corresponding counts are 1,077, 761 and 316; raising that anticipated-path criterion to 8% and 10% further reduces qualification to 625 and 474 records. Front-loaded and back-loaded paths at the same endpoints widen the lower-rule range to 849–1,482 records (Fig. 2d). Thus, the static 1,809-record case is a rule-isolation counterfactual, not a prediction of investor behaviour.

Physical and temporal assumptions alter exact membership more than the separation. Daily peak-shaving, annual peak-shaving and proportional allocation yield 1,809, 475 and 5,544 lower-rule records and 710, 329 and 390 strict-marginal records, respectively (Supplementary Fig. S1a–c). A separate six-weather-year replay yields 2,315–2,682 lower-rule and 1,080–1,321 strict-marginal records on a common grid; none of its strict cohorts reaches 6.5% under the linear 18 CNY kg^{-1} path. Adding the exact 1-MW boundary and adaptive local search reproduces the final 1,809, 1,099 and 710 memberships (Supplementary Fig. S1e). Exact counts are reconstruction-dependent; persistence of the return separation is the cross-reconstruction result.

Replacement access is insufficient to convert sector learning into durable incumbent returns

Low-cost operation delays access to later learning. Across 710 strict-marginal records, the median 6.5% NPV shortfall is 18.5% of initial capital and 30-year operation is 54,800 hours. With a 60,000-hour initial stack life, 701 never replace; only nine do so in 2052–2055 (Fig. 3a). Positive learning gain and replacement coincide because cost, efficiency and life improvements enter only then. Among these nine, the median gain is 2.0% of initial capital and closes 39% of the initial gap; only three cross 6.5% at a flat 28 CNY kg^{-1} (Fig. 3c). Sparse use blocks access, and the payoff usually remains insufficient.

Fixed-cadence tests separate access from sufficiency. All 710 records trigger replacement at 20,000–50,000 hours, yet only four or five meet 6.5% under central learning and five to eight under the source-optimistic path (Fig. 3e). At 60,000 hours, nine trigger and three meet 6.5%; at 80,000–100,000 hours none triggers, and three pass only because they avoid replacement expenditure. Even a 90% stack-cost learning rate combined with a 20,000-hour cadence closes only 39 gaps, leaving 671 unresolved. Cohort-wide reversal requires twenty-fold joint operating improvements and replacement every 20,000–30,000 hours; at 60,000 hours only six upgrade. Across 55 lower-to-higher return comparisons, all five cohorts with median initial gaps below 2% of CAPEX upgrade at rates of 22.8–72.5%, whereas the 24 cohorts with gaps of at least 10% average 0.43% and never exceed 2.66% (Fig. 3f). Reversal is therefore possible, but only where the initial shortfall is already commensurate with the limited gain that can reach an incumbent.

Component incidence further limits how much sector learning can enter an existing plant. The central decomposition assigns 67.5% of installed cost to direct CAPEX and 25% of direct CAPEX to the stack. On that basis, the replaceable stack embodies 18.2% of the capital saving achieved by a 2060 new build; crossing source-grounded component shares and three conditional learning paths gives a range of 11.3–42.3% (Fig. 3b). We then credit the incumbent, at its first replacement, with up to 100% of the contemporaneously available non-stack saving without tax, retrofit cost or repetition. Under a flat 28 CNY kg^{-1} price, this deliberately favourable transfer raises the number crossing 6.5% from three to eight; across the full component boundary the range is five to eight, leaving at least 702 records unresolved (Supplementary Note 5). Price decline is therefore not required for the vintage separation.

Price convergence dominates post-entry survival. At a 2060 endpoint of 22 CNY kg^{-1}, front-loaded, linear and back-loaded paths leave 4, 45 and 298 strict-marginal records above the lower criterion; none reaches 6.5%. At 18 CNY kg^{-1}, the corresponding counts are 0, 0 and 102. The linear-path terminal price required for a strict-marginal record to reach 6.5% has a median of 38.2 CNY kg^{-1} and a 5th–95th percentile range of 33.1–42.2 CNY kg^{-1} (Fig. 3d). Relative to the flat-price, no-learning counterfactual, a linear path to 18 CNY kg^{-1} removes 5.31 billion CNY from cohort NPV at the lower criterion, whereas operating learning restores only 4.7 million CNY, or 0.09% of that loss (Supplementary Note 5). Vintage incidence explains why incumbents cannot capture enough compensation even at constant price; conditional price convergence determines the magnitude of subsequent loss. This mechanism conclusion uses neither scenario-grid frequencies nor the declared-prior audit reported in Supplementary Note 11; the latter is retained only to show prior sensitivity and is not interpreted as an empirical investment probability.

Price-informed screening and capacity flexibility move risk before capital lock-in

Separating expectations at commitment from conditions realised afterwards changes the apparent value of a lower return rule. If the static 28 CNY kg^{-1} lower-rule capacities are committed and the producer price subsequently follows a linear path to 18 CNY kg^{-1}, only 274 of 1,809 records remain above 6.5%; 44.0 of 60.7 billion CNY falls below that criterion and durable output is 0.103 Mt H_2 yr^{-1}. If the same price and operating-learning path is anticipated at commitment, the lower rule qualifies 1,077 records rather than 1,809. Requiring 6.5% under that anticipated path retains 761 records, 18.4 billion CNY and 0.120 Mt yr^{-1} (Fig. 4b). Relative to static lower-rule entry, this screen commits 70% less capital while retaining 16% more output that remains above 6.5% under the realised path.

A higher static criterion alone does not substitute for forward information. The static 6.5% rule selects 1,099 records, but only 490 remain above 6.5% under the linear 18 CNY kg^{-1} path and 10.4 of 33.1 billion CNY becomes exposed. A screen robust to all three timing shapes at the 18 CNY kg^{-1} endpoint retains 408 records, 8.6 billion CNY and 0.060 Mt yr^{-1}; at 22 CNY kg^{-1}, the corresponding linear and timing-robust screens retain 971 and 781 records (Fig. 4a,f). Zero non-durable exposure is imposed by these screens, so the scientifically informative quantities are the capital and supply that survive each information requirement, not the zero itself.

Pre-investment capacity flexibility converts resource error into avoided commitment rather than post-investment loss. Holding the static 28 CNY kg^{-1} producer price and no-learning conditions used at entry fixed, 75% realisation of the low-cost electricity assumed at design leaves 157 locked-capacity records and 1.18 billion CNY below the lower criterion while producing 0.292 Mt H_2 yr^{-1}; 325 records, contributing 0.105 Mt yr^{-1}, reach 6.5%. Under continuous downsizing without a minimum-build cutoff, each additional 25 percentage points of adjustability leaves the same 3.93 billion CNY of planned CAPEX uncommitted. The first increment reduces exposure to 99 records and 0.61 billion CNY while retaining 0.280 Mt yr^{-1}, of which 0.145 Mt comes from records reaching 6.5%; at 50% adjustability, exposure is eliminated and 7.86 billion CNY, or 12.9% of initially selected capital, remains uncommitted. Although total output then falls only to 0.267 $\mathrm{Mt\,yr^{-1}}$, 91.3% of the locked outcome, output from the 656 records reaching 6.5% rises from 0.105 to 0.169 $\mathrm{Mt\,yr^{-1}}$, 60% above the locked case. Across the four equal capital increments, additional records reaching 6.5% decline from 178 to 127, reducing the marginal durability yield from 45.3 to 32.3 records per CNY billion not committed (Fig. 4e,g). Flexibility therefore sheds capital faster than production. Under the primary 1-MW engineering convention, 25% adjustability instead maps 154 sub-threshold capacities to no-build and mechanically reports zero exposure (Fig. 4d). The option value is robust, but the adjustment share at which exposure reaches zero depends on minimum viable project scale and is not a universal 25% threshold.

Retrospective support is much larger than the illustrative policy limits tested. Under the linear 18 $\mathrm{CNY\,kg^{-1}}$ path, no strict-marginal record is upgraded by either a 4 $\mathrm{CNY\,kg^{-1}}$ 15-year producer premium or a 25% upfront capital grant. The median requirements are 8.8 $\mathrm{CNY\,kg^{-1}}$ and 44.5%, and even the least costly record requires more than the two illustrative limits (Fig. 4c). These are plant-gate financing gaps: a China-specific delivery-cost transfer replaces the strict cohort entirely, and the delivery-exposed prior produces no upgrade. Route and offtake screening must therefore precede instrument design. Forward screening and capacity flexibility act on the source of production-side exposure, whereas ex post support finances a return gap after the capital configuration has been fixed.

Discussion

The curtailment-learning trap refines the vintage-capital test from whether a component is replaceable to whether replacement occurs within the economic life of the asset. A replaceable stack is only a potential conduit for embodied technical change. In the reference cohort, sparse operation leaves 701 of 710 strict-marginal records below the 60,000-h trigger; among the nine that replace, central learning closes only three return gaps at constant price. Forcing all 710 records to replace every 20,000–50,000 hours removes the access constraint, yet central learning still raises only four or five above 6.5%, and a source-optimistic path raises five to eight. Even a 90% stack-cost learning rate with 20,000-h replacement closes only 39 gaps. Replaceability is therefore necessary but not sufficient: utilisation first governs access, and the limited share of learning embodied in the replacement then governs sufficiency. Green hydrogen is not an exception to embodied technical change, but a sharp case in which the operational use of a component determines whether its new vintage ever arrives.

This distinction shifts effective de-risking towards information available before capital becomes irreversible[39, 40]. Capacity support does not ensure that incumbent stacks accumulate the hours needed to receive later improvements. Verifiable low-cost operating-hour commitments, through long-term power contracts, guaranteed operating windows or stable loads, are therefore better aligned with the mechanism than installed capacity alone. Under a linear price path to 18 $\mathrm{CNY\,kg^{-1}}$, a path-informed 6.5% screen commits 70% less capital than static lower-rule entry while retaining 16% more output above the comparator. When 75% of designed low-cost electricity is realised, continuous 50% pre-investment adjustment withholds 7.86 billion CNY and retains 91% of locked-case output, while increasing output from configurations above 6.5% by 60%. These results favour credible utilisation and capacity flexibility before final investment decision, rather than assuming later deployment repairs a low-use asset.

The non-random inventory and reconstructed hourly allocation yield conditional exposure, not national potential, although six ERA5 replays preserve resource ranking. Hydrogen is valued at the plant gate with full sales; transport, storage, local demand and basin-level water constraints can reduce feasible membership, especially in Xinjiang and Qinghai[41, 42]. Buffering or transmission could raise utilisation and reopen the replacement channel, while different degradation physics could weaken the link between hours and replacement. The firm rules are screening scenarios rather than national policy or estimates of equity cost, and 6.5% is a comparator rather than a break-even condition. The mechanism remains: sector learning reaches an incumbent only through changes within its own operating and replacement history.

Methods

Research design and system boundary

The analysis unit is an operating wind or utility-scale photovoltaic `ObjectId` record in the project inventory. Resource reconstruction, capacity candidates, nominal equity cash flow, entry and durability are evaluated at record level before provincial and inventory-level aggregation. Separate construction phases of one named facility can appear as multiple records, so all counts are described as project records rather than distinct physical plants. The primary financial horizon contains 30 operating years, from 2026 through 2055; 15-, 20-, 25- and 35-year horizons are sensitivity cases. Conditional market and learning paths are specified through 2060 to retain a common endpoint, but only values realised through 2055 enter the primary asset cash flow. Continued access to low-cost electricity represents contract renewal, host repowering or replacement supply. The primary branch uses system-constrained electricity. A mutually exclusive full-output branch redirects all modelled renewable output to electrolysis and charges the provincial feed-in price as an opportunity-cost proxy, providing a physical diversion boundary.

The financial boundary includes the installed electrolysis system: electrolyser package, rectification and electrical systems, gas–liquid separation and purification, cooling, water treatment, buildings, engineering, procurement, construction and contingency. It excludes the existing renewable host asset, off-site compression and storage, hydrogen pipelines, distribution and end-use facilities. Construction period, working capital and residual value are zero in the primary boundary, and all hydrogen is sold at the producer side. Separate tests add zero to two construction years with capitalised interest and after-tax residual values of 0–20% of initial capital. These choices define the production-side investment boundary.

Project inventory and hourly low-opportunity-cost electricity

The inventory combines the June 2024 Global Wind Power Tracker and Global Solar Power Tracker releases and contains 10,214 operating records with capacity, technology, coordinates and province[43]. The 2020 hourly reference files are research-group station-level model outputs previously used and documented by Li and Zhang[44], and are reused here; that article describes simulated generation records and reconstructs hourly curtailment by daily peak-shaving to provincial rates. The files contain 8,784 leap-year hours and 15,573 unique identifiers, of which all 10,214 inventory identifiers match one-to-one. We sum the grid-delivered and constrained components only to recover a pre-allocation full-output shape; neither the earlier study's projected 2030 curtailment rates nor its hourly constrained quantities enter our annual calibration. We instead calibrate annual constrained energy to 2025 province- and technology-specific utilisation rates[45]. A separate 42-cell comparison with public monthly monitoring data supports seasonal utilisation magnitudes but does not identify hourly event timing[46]. For province p and technology s, the annual constrained share is $r_{ps} = 1 - U_{ps}$. The reference daily peak-shaving reconstruction solves a threshold θ_{id} for record i and day d:

$$r_{ps} = 1 - U_{ps}, \qquad \sum_{h=1}^{24} \max(P_{idh} - \theta_{id}, 0) = r_{p(i)s(i)} \sum_{h=1}^{24} P_{idh}, \tag{1}$$

where P_{idh} is calibrated full output. The reference reconstruction applies r_{ps} uniformly to records within each province–technology group; record-level allocation is treated as partially identified.

Two concentration extremes therefore allocate the same group total greedily to records in ascending or descending installed-capacity order, capped by each record's full output. All three allocations conserve every group total exactly. The hourly proxy is $C_{idh} = \max(P_{idh} - \theta_{id}, 0)$; thresholds are solved by bisection and annual energy is rescaled exactly. Annual peak-shaving uses one threshold over the annual duration curve, whereas proportional allocation assigns the constrained share at every hour. The resulting inputs are modelled generation and allocation profiles used to test the investment mechanism.

Weather-year sensitivity uses ERA5 hourly 100-m wind components, 2-m temperature, surface pressure and surface solar radiation downwards for 2020–2025[47, 48]. Variables are bilinearly interpolated to each record. Wind output uses an air-density-corrected generic power curve, and photovoltaic output uses a temperature-corrected irradiance model. Each record is calibrated to its 2020 reference equivalent full-load hours; this scalar is fixed in later weather years so that replay changes meteorological shape while holding the 2025 utilisation-rate calibration constant.

Figure 1a separates the analytical main map from its locator inset. The main-map land mask, coastline and visible national and provincial linework use OpenStreetMap coastline polygons and administrative relations[49]. Records and all main-map layers use the same China Albers equal-area projection. The boxed South China Sea locator is a cleaned display derivative of the Ministry of Natural Resources standard-map product GS(2023)2767[50]; it is not used for masking, spatial assignment or any numerical result. Both sources provide cartographic context rather than legal adjudication. Query timestamps, product identifiers, transformations and integrity hashes are archived with the figure data.

Electrolyser dispatch and capacity choice

For candidate electrolyser capacity K, the plant operates only when available low-opportunity-cost power exceeds the minimum-load fraction λK. Absorbed power in hour t is

$$A_{it}(K) = \mathbb{1}(C_{it} \geq \lambda K) \min(C_{it}, K). \tag{2}$$

Annual absorbed electricity, operating hours and hydrogen output are

$$\begin{aligned} E_i(K) &= \sum_t A_{it}(K)\Delta t, \\ h_i(K) &= \sum_t \mathbb{1}(A_{it} > 0)\, \Delta t, \\ H_{iy}(K) &= 1000 \sum_t \frac{A_{it}(K)\Delta t}{\varepsilon_{iyt}}. \end{aligned} \tag{3}$$

Here $\Delta t = 1\,\mathrm{h}$, E_i is measured in MWh, h_i in hours and H_{iy} in kg; the factor 1000 converts MWh to kWh. The alkaline main case uses $\lambda = 0.30$. Candidate capacities are generated from 128 nested resource-capture fractions, with the exact 1-MW engineering lower bound added as an independent candidate. Candidates below 1 MW or with zero production are removed. For return criterion r, every record is independently resized by

$$K_i^*(r) = \underset{K \in \mathcal{K}_i,\, K \geq 1\,\mathrm{MW}}{\arg\max} \ \mathrm{NPV}_i(K; r). \tag{4}$$

A strict-marginal record has at least one capacity with non-negative NPV at the low criterion and no capacity with non-negative NPV at 6.5%. Grid density, the 1-MW boundary, continuous local search, lower minimum loads and proton-exchange membrane (PEM) operation are assessed in the Supplementary Information. Dispatch is directly coupled without a battery or hydrogen buffer, so the minimum-load and oversizing results describe the unbuffered operating configuration.

Installed-system cost, nominal equity cash flow and return criteria

For capacity K_i in MW and installed-system unit cost c_{sys} in CNY kW^{-1}, initial investment, debt and equity are

$$I_i = 1000 K_i c_{\text{sys}}, \qquad D_{i0} = \delta I_i, \qquad Eq_{i0} = (1-\delta) I_i, \tag{5}$$

where δ is the debt share. The World Bank reports 500–1,500 USD kW^{-1} for installed alkaline systems; using a transparent conversion assumption of 7.2 CNY per USD gives deterministic bounds of 3,600 and 10,800 CNY kW^{-1}, with 7,200 CNY kW^{-1} as the midpoint central case[26]. The IEA's 600–1,200 USD kW^{-1} China interval provides contextual equipment-cost evidence, while the World Bank interval defines the installed-system boundary used here[51]. Fixed operations and maintenance (O&M) is 5% of initial capital per year, and an explicit stack replacement costs 11% when triggered[25]. Mutually exclusive all-in operating-cost conventions are tested to avoid double counting[26]. Constrained electricity costs 0.10 CNY kWh^{-1} centrally and 0–0.20 CNY kWh^{-1} in sensitivity. Water uses provincial values reported in Supplementary Table 3 of the earlier study[44] and 15 kg of water per kg H_2 centrally, with 10–60 kg kg^{-1} tested as a direct-cost boundary.

All model price and cost scenarios are expressed in constant 2026 CNY and converted to nominal cash flow using a common escalation rate π. The 28 CNY kg^{-1} entry reference is anchored numerically to the December 2024 cross-route producer sample:

$$X_y^{\text{nom}} = X_y^{\text{real}} (1+\pi)^{y-2026}, \qquad R_{iy} = H_{iy} P_y (1+\pi)^{y-2026}. \tag{6}$$

The central accounting case uses $\pi = 2\%$ and tests 0%, 1% and 3%. Nominal loan rates and nominal return criteria are applied only to nominal cash flows. The central financing structure is 70% debt, a 3.5% nominal loan rate, a 15-year term and two years of principal grace. Corporate income tax is 25% and tax losses can be carried forward for five years[52]; a ten-year straight-line depreciation schedule is a transparent accounting assumption applied to initial and replacement capital.

Annual free cash flow to equity subtracts operations, stack replacement, tax, interest and principal from revenue. Classification is based on NPV consistency because replacement expenditure can create multiple cash-flow sign changes:

$$\begin{aligned} CF_{iy}^{Eq} &= R_{iy} - O_{iy} - M_{iy} - T_{iy} - Int_{iy} - Prin_{iy}, \\ \text{NPV}_i(K; r) &= -Eq_{i0} + \sum_{y=1}^{Y} \frac{CF_{iy}^{Eq}}{(1+r)^y}. \end{aligned} \tag{7}$$

A record passes criterion r when the NPV of its criterion-specific optimum is non-negative. "Durable 6.5%" means that the locked 2026 investment and capacity produce non-negative 30-year equity NPV at both the lower rule and 6.5%. Capacity remains fixed after construction, and classification uses equity value without debt-service-coverage, reserve-account or default tests. It is therefore a return-screen classification, not a bankability test or prediction of actual investment.

We operationalise CHN Energy Changyuan's 2026 rule with the mean nominal five-year ChinaBond government bond yield over 20 trading days from 4 June to 2 July 2026, reported as approximately 1.45%[33, 34]. "Five-year" denotes bond maturity, and the unrounded arithmetic mean is retained in the machine-readable calculation. The 6.5% comparator comes from Huadian Energy's separate 2025 rule for hydrogen-based energy[35]. These publicly disclosed firm-level criteria define an institutional scenario comparison; the lower value is not treated as a risk-adjusted cost of equity. We additionally scan nominal equity criteria from 1 to 10% in 0.25-percentage-point increments.

To test whether the learning result is mechanically tied to either firm rule, we construct a dense return-screen surface under the flat 28 CNY kg^{-1} reference. Eleven nominal criteria (approximately 1.45%, 2%, 3%, 4%, 5%, 6%, 6.5%, 7%, 8%, 9% and 10%) generate all 55 lower-to-higher comparisons. Capacities are selected under each lower criterion without crediting future operating improvement, locked, and then revalued after central operating learning at every higher criterion. None of these pairs is interpreted as a representative weighted average cost of capital.

Vintage-specific operating learning and price paths

The initial alkaline-stack electricity use is $\varepsilon_0 = 55\,\mathrm{kWh\,kg^{-1}\,H_2}$ and stack life is 60,000 operating hours. Within-stack degradation is represented by

$$\varepsilon(a) = \varepsilon_0(1 + \gamma a), \qquad \gamma = \frac{2(57.3/55 - 1)}{60000}, \tag{8}$$

so average lifetime electricity use is $57.3\,\mathrm{kWh\,kg^{-1}\,H_2}$. An operating asset receives improved efficiency, life and replacement cost only when a stack replacement is triggered by cumulative operating hours. Manufacturing learning in complete systems and balance-of-plant/engineering affects future new investment and is never used to reduce the 2026 sunk capital or annual fixed O&M of an existing plant.

The stack-cost experience curve is

$$c_{\mathrm{stack},t} = c_{\mathrm{stack},0}\left(\frac{Q_t}{Q_0}\right)^{-b}, \qquad b = \frac{-\ln(1 - LR)}{\ln 2}, \tag{9}$$

where Q_t is cumulative deployment and LR is the learning rate. The path starts at 4 GW, a rounded end-2025 global installed-capacity observation[53]. The subsequent 140, 700, 1,400 and 2,200 GW anchors and the 13% equipment/stack and 5% balance-of-plant experience rates define an illustrative central calendar mapping, not fitted parameters or forecasts; Supplementary Table S4 distinguishes external benchmarks from every author-constructed node. They determine timing in the labelled central case but do not identify the mechanism: replacement access is calculated from each record's cumulative operating hours before future learning is credited, while sufficiency is tested separately with unfloored 0–90% learning rates and fixed 20,000–100,000-h cadences. A 20-GW starting normaliser provides an additional anchor sensitivity. Electricity use and stack life are bounded trajectories interpolated in log-deployment space: they move from 55 to $50\,\mathrm{kWh\,kg^{-1}\,H_2}$ and from 60,000 to 90,000 h by 2060[23, 25, 26, 51]. The central replacement-stack cost factor reaches its imposed 0.55 floor in 2028 and remains there; an existing record obtains the applicable factor only in its actual replacement year.

Three incidence and falsification boundaries test whether the result follows from restricted learning assumptions. The World Bank distinguishes direct and indirect CAPEX, reports 65:35–70:30 direct-to-indirect shares for large greenfield projects and places the stack at 20–50% of direct CAPEX[26]. We therefore use $d \in \{0.50, 0.675, 0.70\}$ and $q \in \{0.20, 0.25, 0.35, 0.50\}$, with central values $d = 0.675$ and $q = 0.25$. Installed capital is decomposed into stack ($s_S = dq$), other direct equipment ($d - s_S$) and indirect balance-of-plant/engineering ($1 - d$). The central $s_S = 0.16875$ is a component-accounting share; the separate 11% replacement-event expenditure remains a cash-flow assumption. The share of new-build capital savings embodied in the replaceable stack is

$$\phi_t = \frac{s_S(1 - f_t^S)}{s_S(1 - f_t^S) + (d - s_S)(1 - f_t^E) + (1 - d)(1 - f_t^B)}, \tag{10}$$

where f_t^S, f_t^E and f_t^B are the stack, other direct-equipment and indirect-capital cost factors. Under the central path, $\phi_{2060} = 0.182$; the full component-share and learning-path boundary is

0.113–0.423. A deliberately favourable transfer boundary then credits an incumbent at its first stack replacement with

$$T_{it} = \lambda I_{i0} \left[(d - s_S)(1 - f_t^E) + (1 - d)(1 - f_t^B) \right], \qquad \lambda \in [0, 1], \tag{11}$$

where I_{i0} is initial installed-system CAPEX and λ is evaluated in 0.01 increments. The transfer is added directly to equity cash flow without tax, retrofit cost or repetition, making it an upper bound rather than a proposed instrument. A reduced-form pass-through boundary with $\beta \in \{0, 0.25, 0.50, 0.75, 1\}$ is retained only as an auxiliary SI diagnostic linking future new-build cost decline to producer price. Because β is not estimated from market data, no central value is assigned and this diagnostic is excluded from headline counts and mechanism inference. A second auxiliary test scales incumbent access to electricity-use, stack-life and replacement-cost improvements from zero to the central path.

The stack-learning falsification boundary removes the 0.55 replacement-cost floor and scans the stack-cost learning rate from 0 to 90% per doubling in one-percentage-point increments. It combines the central endogenous lifetime path with fixed replacement cadences of 20,000, 40,000, 60,000, 80,000 and 100,000 operating hours. The event replacement expenditure is separately varied from 6 to 45% of installed capital. These tests ask how extreme learning strength, replacement frequency and replacement scope must become before the strict-marginal conclusion reverses; they are not forecasts of rational replacement policy.

Long-run hydrogen prices are conditional paths. For terminal price P_{2060} and timing function g_s, the constant-2026-CNY price is

$$P_y = P_{2026} + (P_{2060} - P_{2026}) g_s(\tau_y), \qquad \tau_y = \frac{y - 2026}{2060 - 2026}, \qquad g_s \in \{\sqrt{\tau}, \tau, \tau^2\}, \tag{12}$$

where the three functions denote front-loaded, linear and back-loaded convergence. The initial price is 28 $\mathrm{CNY\,kg^{-1}}$ and terminal values are 22, 18, 15 and 12 $\mathrm{CNY\,kg^{-1}}$. The endpoint is a conditional 2060 market anchor rather than the asset's terminal-year sale price; the primary project receives the corresponding path values only through 2055. Paired counterfactuals identify price and operating-learning contributions:

$$\begin{aligned} \Delta \mathrm{NPV}_{\mathrm{price}} &= \mathrm{NPV}(P_s, L_0) - \mathrm{NPV}(P_0, L_0), \\ \Delta \mathrm{NPV}_{\mathrm{learn}} &= \mathrm{NPV}(P_s, L) - \mathrm{NPV}(P_s, L_0). \end{aligned} \tag{13}$$

We distinguish information available at commitment from subsequent realisation. The static reference uses a constant 28 $\mathrm{CNY\,kg^{-1}}$ price and credits no future operating learning when capacity is selected; it isolates the effect of changing the return criterion under common contemporaneous assumptions. Twelve anticipated cases cross the four terminal prices and three timing shapes and allow the replacement-mediated operating path to enter the commitment calculation. An expectation–realisation matrix then evaluates capacities selected under static 28, anticipated 22-linear and anticipated 18-linear conditions against realised flat-28, 22-linear and 18-linear paths. No probability is assigned to these cases.

The reference lifetime calculation holds low-cost electricity equivalent hours at their 2025-calibrated level. A persistence audit scales annual active hours, absorbed electricity and electricity expenditure by a linear factor from 1 in 2026 to 0.50, 0.75, 1.00 or 1.25 in 2060, spanning declining and expanding contract-equivalent access.

Forward screening, capacity flexibility and support boundaries

Forward screening places a long-run price endpoint and timing in the capacity-choice problem before FID and retains only capacities with non-negative NPV at both criteria. Timing-robust screening first selects the engineering-eligible capacity with the greatest worst-case 6.5% NPV across $\mathcal{S}$, the front-loaded, linear and back-loaded paths:

$$K_i^{\text{rob}} = \arg\max_{K \in \mathcal{K}_i^{\text{elig}}} \min_{s \in \mathcal{S}} \text{NPV}_i(K; 6.5\%, s), \tag{14}$$

$$S_i^{\text{rob}} = \mathbb{1}\left[\min_{s \in \mathcal{S}} \text{NPV}_i(K_i^{\text{rob}}; 6.5\%, s) \geq 0\right] \mathbb{1}\left[\min_{s \in \mathcal{S}} \text{NPV}_i(K_i^{\text{rob}}; r_{\text{low}}, s) \geq 0\right]. \tag{15}$$

Here $\mathcal{K}_i^{\text{elig}}$ is the engineering-eligible capacity set. A selected record is retained only when $S_i^{\text{rob}} = 1$, matching the implemented sequence of capacity selection followed by the two-criterion worst-case screen. A stricter minimax screen replaces $\mathcal{S}$ with the Cartesian product of the three timing shapes and terminal prices from 12 to 22 $\text{CNY}\,\text{kg}^{-1}$ in 1-CNY increments. Because project NPV is monotone in price, this endpoint-and-timing screen is governed by the 12 $\text{CNY}\,\text{kg}^{-1}$ boundary but is reported separately to make the information assumption explicit.

Capacity flexibility describes the share that can be adjusted before construction after a resource deviation is known. If K_i^0 is original capacity and $K_i^*(\rho)$ is the re-optimised capacity under resource realisation ρ, installed capacity at adjustability a is

$$K_i(a, \rho) = K_i^0 + a\left[K_i^*(\rho) - K_i^0\right], \qquad a \in \{0, 0.25, 0.50, 0.75, 1\}. \tag{16}$$

Among positive engineering-eligible capacities, $K_i^*(\rho)$ maximises low-criterion NPV. If none has non-negative NPV, $K_i^*(\rho) = 0$, representing cancellation before commitment; the primary engineering convention also maps interpolated capacities below 1 MW to zero. Because this rule can create a discontinuity, a separate M129 audit repeats the 75% resource-realisation surface with minimum build sizes of 0, 0.1, 0.5, 1 and 2 MW; the zero-bound case permits continuous downsizing and distinguishes economic exposure from threshold cancellation. Exposed capital is gross initial investment in positive-capacity records that cease to satisfy the low-return criterion after the resource deviation. Avoided capital is planned investment withheld through downsizing or cancellation. Complete flexibility assumes that resource error is resolved before FID and that capacity can be adjusted without cancellation, lead-time or interconnection costs. Support boundaries search for the minimum constant 2026–2040 producer-price premium or upfront capital grant that makes a locked strict-marginal record satisfy 6.5%, yielding equivalent financing gaps for comparison with forward screening.

Robustness, uncertainty and quality control

Deterministic robustness spans installed-system capital cost, constrained-electricity price, operating cost, resource realisation and persistence, debt share, nominal loan rate, escalation, evaluation horizon, minimum load, alkaline and PEM technology, learning intensity, learning-start anchor, terminal hydrogen price, construction duration, residual value, producer-side netback and electrical buffering. The constrained-electricity branch contains 972 factorial combinations, and a mutually exclusive full-output branch adds 324 opportunity-cost combinations, for 1,296 deterministic cases in total. All combinations have equal diagnostic status and define a robustness domain; their frequencies are not probabilities. Because empirical joint distributions are unavailable, a separate prior-sensitivity audit declares six alternative priors over terminal price, timing, operating learning, weather-year resource factors, price–learning dependence and delivery netback and uses 5,000 draws per prior case. It reports consequences of declared assumptions only; it neither estimates historical project probabilities nor validates the headline mechanism. Within each province–technology group, two annual-energy-conserving concentration extremes assign constrained energy first to smaller or larger records, respectively, partially identifying outcomes under the unknown allocation. The financing contribution to within-grid dispersion is evaluated over its specified parameter range. Netback penalties of 0–4 CNY kg^{-1} are applied before reoptimising entry. Electrical buffers of zero to four equivalent hours are dispatched against the original hourly profiles; free lossless storage defines a physical upper bound, whereas costed cases include initial capital, fixed operation and maintenance, round-trip loss and periodic replacement.

Capacity discretisation is checked against 128- and 256-candidate nested grids and continuous local search. Weather shape is checked by replaying ERA5 2020–2025. In addition to the M129 resource check, the complete entry, learning and support calculations are replayed with a common 16-candidate grid for each weather year; these G16 counts are reported as a separate robustness experiment and are never pooled with the M129 main cohort. Set stability is measured by

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}. \tag{17}$$

All figures and reported values are generated from record-level output tables. Automated checks cover annual energy conservation, cohort denominators, criterion nesting, nominal cash-flow reconstruction, capacity-search convergence and figure–text consistency. Provincial water price and constrained-electricity settlement are retained as transparent boundaries where evidence is insufficient for causal or implementation claims.

Resource availability

Lead contact

Further information and requests for resources should be directed to the lead contact, Haoran Zhang (h.zhang@pku.edu.cn).

Materials availability

This study did not generate new physical materials.

Data and code availability

Data.

Redistributable project-inventory fields, annual utilisation inputs, processed ERA5 outputs, parameter-source registers, figure source data and result tables are maintained in the accompanying public GitHub repository at github.com/1223044298-ship-it/curtailment-traps-early-green-hydrogen-assets-beyond-the-reach-of-learning. The two 10,214-by-8,784 `float32` hourly profile arrays used by the analysis are openly available as versioned assets in the hourly-inputs-v1.0.0 GitHub Release. Their dimensions, storage order, byte counts, SHA-256 digests and direct download URLs are recorded in `analysis_code/HOURLY_INPUTS_SHA256.csv`; the supplied downloader places them at the paths expected by the workflow and rejects files that fail validation. These arrays enable a clean rerun from analysis-ready hourly inputs through the reported results. They are deterministic derivatives of 2020 monthly model-output files generated within the authors' research group, previously used and documented by Li and Zhang[44], and reused here rather than generated specifically for this study. Their provenance is fixed to that article and its public source-code repository at commit 43af9fe; that code reads, but does not distribute or regenerate, the upstream monthly files. Those upstream files are not redistributed and remain available from the corresponding author at h.zhang@pku.edu.cn upon reasonable request, subject to contributor and institutional approval. The archive therefore does not claim independent regeneration of the public arrays from the least-processed monthly source files. These files are model outputs and are not represented as independently observed station dispatch. Source ERA5 variables are publicly available from the Copernicus Climate Data Store under its applicable terms. The manuscript-aligned revision is identified by a fixed Git commit; the accepted release will be archived under a versioned DOI before publication.

Code.

Custom code for constrained-electricity reconstruction, ERA5 replay, electrolyser dispatch and capacity optimisation, nominal equity cash flow, learning-flip boundaries, forward screening, capacity flexibility and figure generation is maintained in the accompanying GitHub repository. The submission snapshot includes the software environment, execution order, input inventory and SHA-256 manifests. Running `python analysis_code/download_hourly_inputs.py` retrieves and verifies the public hourly arrays; together with the packaged inputs, they support an analysis-ready-input-to-results rerun and regeneration of all figures. Regenerating those arrays from the upstream monthly model outputs remains a separate request-access step documented in `analysis_code/INPUTS_REQUIRED.csv`. No archival DOI has yet been assigned; the accepted public release will receive a versioned archival DOI before publication.

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Author contributions

Jinwei Liang: Conceptualization, Methodology, Visualization, Writing – original draft. Zhiling Guo: Writing – review & editing. Haoran Zhang: Conceptualization, Methodology, Writing – review & editing, Supervision.

Declaration of interests

The authors declare no competing interests.

Supplemental information

Supplementary information accompanies this manuscript. Correspondence and requests for materials should be addressed to Haoran Zhang (h.zhang@pku.edu.cn).

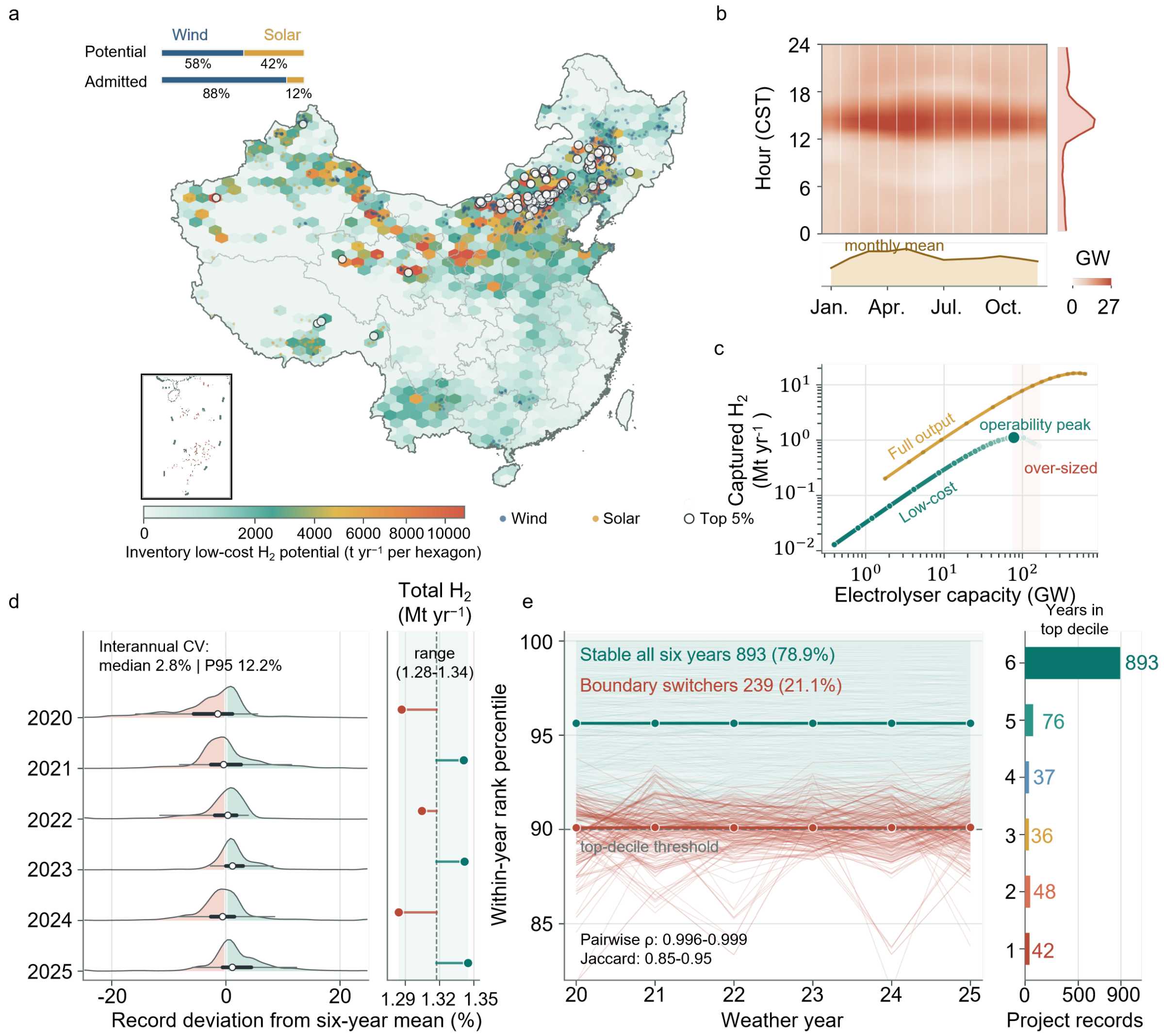

Figure 1: Spatial, temporal and operating constraints delimit hydrogen production from low-opportunity-cost electricity within the project-record inventory. **a**, Spatial distribution of hydrogen potential from system-constrained electricity; dark records meet the lower return criterion, open circles identify the top 5% of admitted records ranked by annual hydrogen output, and stacked bars compare the wind–solar composition of physical potential and admitted production. Main-map coastline, land mask and visible administrative linework derive from OpenStreetMap (© OpenStreetMap contributors) and share the station projection; the boxed South China Sea locator is a cleaned display derivative of standard-map product GS(2023)2767 and is not used analytically. The map colour scale is capped at the occupied-hexagon 97.5th percentile. **b**, Month–hour climatology with diurnal and seasonal margins; its colour scale is capped at the 98th percentile. **c**, Operating frontier for electrolyser capacity, realised hydrogen and active low-cost hours under the mutually exclusive constrained-electricity and full-output branches. **d**, Record-level deviations from six-year mean hydrogen potential and inventory totals for ERA5 weather years 2020–2025; the annotation reports the across-record distribution of interannual coefficients of variation. Density tails are clipped at ±25% for display, while white points, thick lines and thin lines denote medians, interquartile ranges and 5th–95th percentiles calculated from the unclipped values. **e**, Rank trajectories and persistence among records in the top resource decile.

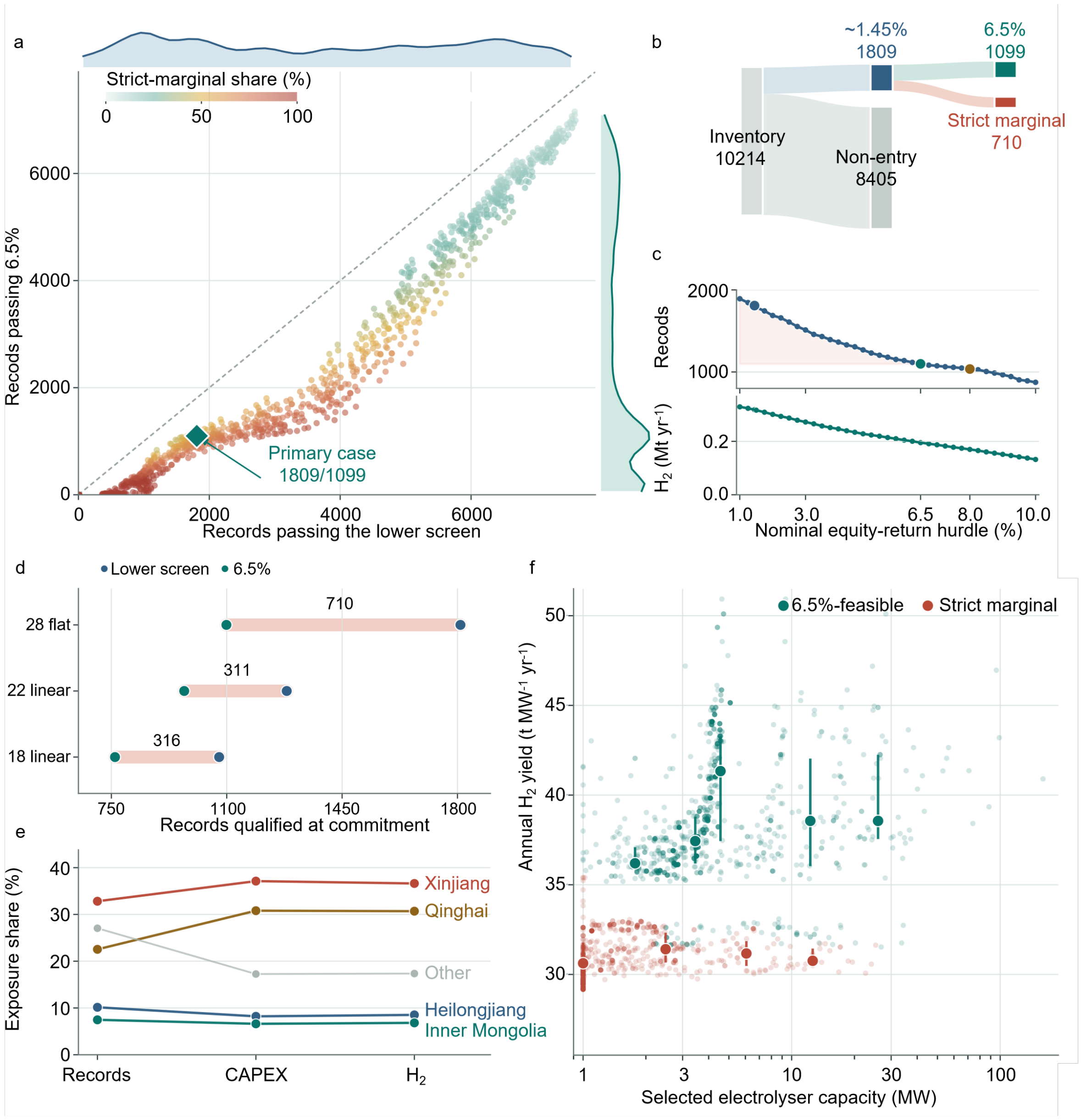

Figure 2: Return criteria create a continuous and information-dependent qualification boundary. **a**, Records passing the lower and 6.5% screens across 972 deterministic constrained-electricity configurations; colour gives the full 0–100% strict-marginal share, marginal curves show the distributions and the diamond marks the primary case. The configurations are an unweighted sensitivity ensemble, not a probability distribution. **b**, Flow from the 10,214-record inventory to static lower-rule qualification, non-qualification, 6.5%-qualified and strict-marginal cohorts. **c**, Qualified records and annual hydrogen after independent resizing across a continuous 1–10% nominal equity-return frontier; highlighted points mark the lower rule, 6.5% and 8%. **d**, Paired qualification under a static 28 CNY kg^{-1} reference and linear 22 and 18 CNY kg^{-1} paths anticipated at commitment; coral spans give the strict-marginal cohort. **e**, Provincial shares of strict-marginal records, gross installed-system capital and annual hydrogen output. **f**, Record-level electrolyser capacity and annual hydrogen yield for 6.5%-feasible and strict-marginal records. Large points and vertical intervals show unconnected medians and interquartile ranges in shared logarithmic bins; only bins containing at least 20 records are summarised.

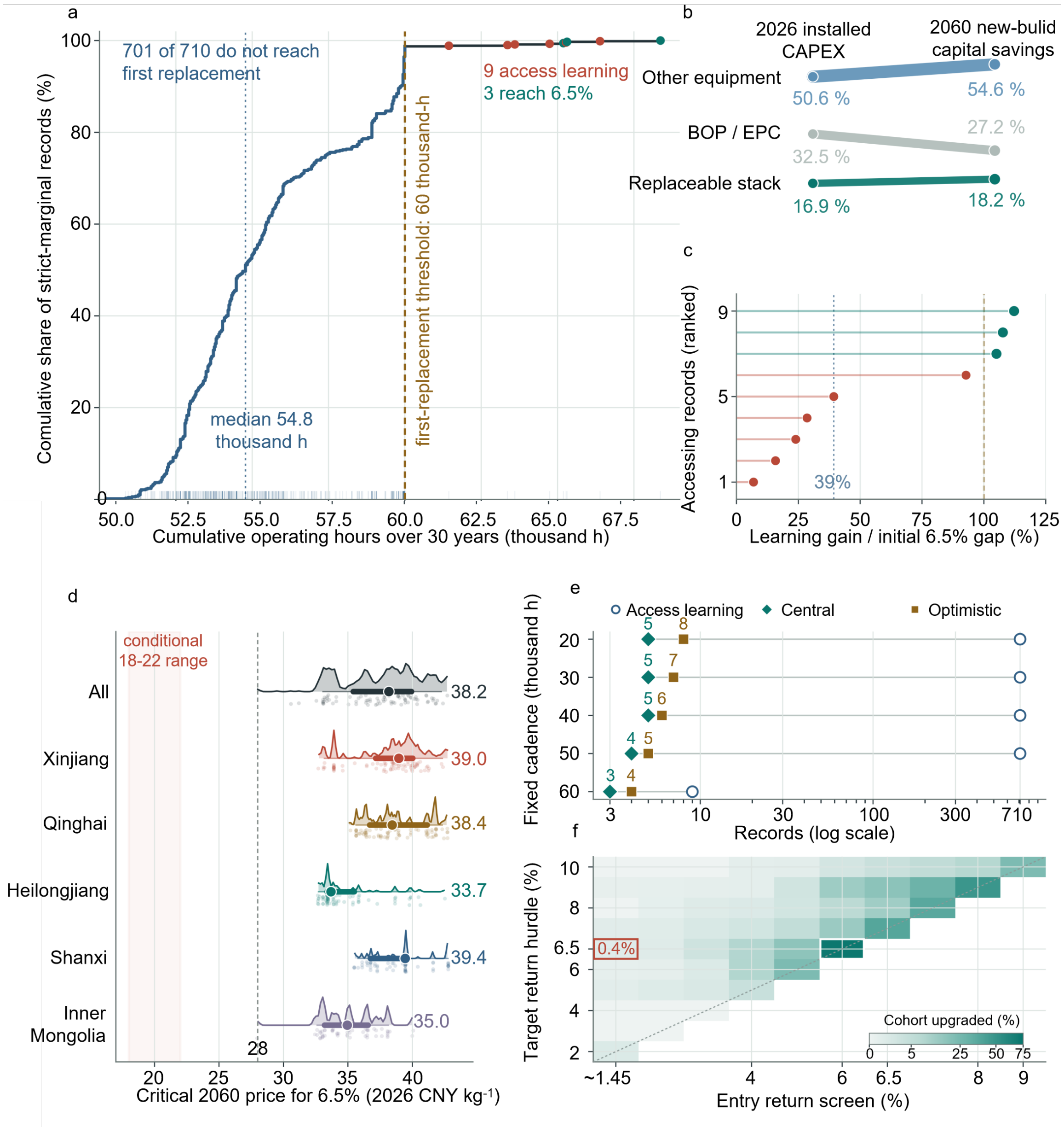

Figure 3: Replacement access is necessary but insufficient for operating learning to repair incumbent returns. **a**, Empirical cumulative distribution of 30-year operating hours across 710 strict-marginal records relative to the central 60,000-hour first-replacement threshold. The blue curve segment identifies records that do not trigger replacement; coral points identify six records that access learning but remain below 6.5%, and teal points identify the three accessing records that close their initial 6.5% NPV gaps. Three near-coincident observations retain their exact coordinates and are layered from left to right using a common display size. **b**, Change in the component shares of 2026 installed CAPEX and 2060 new-build capital savings; only the stack-embodied share is accessible to incumbents, and the displayed source-grounded component boundary spans 11.3–42.3%. **c**, Share of each initial 6.5% NPV gap closed by central operating learning among the nine records that trigger replacement, ranked by closure share; the vertical line at 100% denotes complete closure. **d**, Distributions of the conditional 2060 terminal price required to reach 6.5%, overall and in the principal exposed provinces; points, thick intervals and thin intervals denote medians, interquartile ranges and 5th–95th percentile ranges, respectively. **e**, Fixed-cadence falsification test comparing records that trigger replacement with records reaching 6.5% under central and source-optimistic learning; at 20,000–50,000 hours all 710 records access learning but only four to eight upgrade. **f**, Flat-price return-screen surface across all 55 lower-to-higher criterion pairs. Cell colour gives the share of each locked marginal cohort upgraded by central operating learning; the outlined cells mark the approximately 1.45–6.5% comparison and the narrow 6.0–6.5% counterfactual.

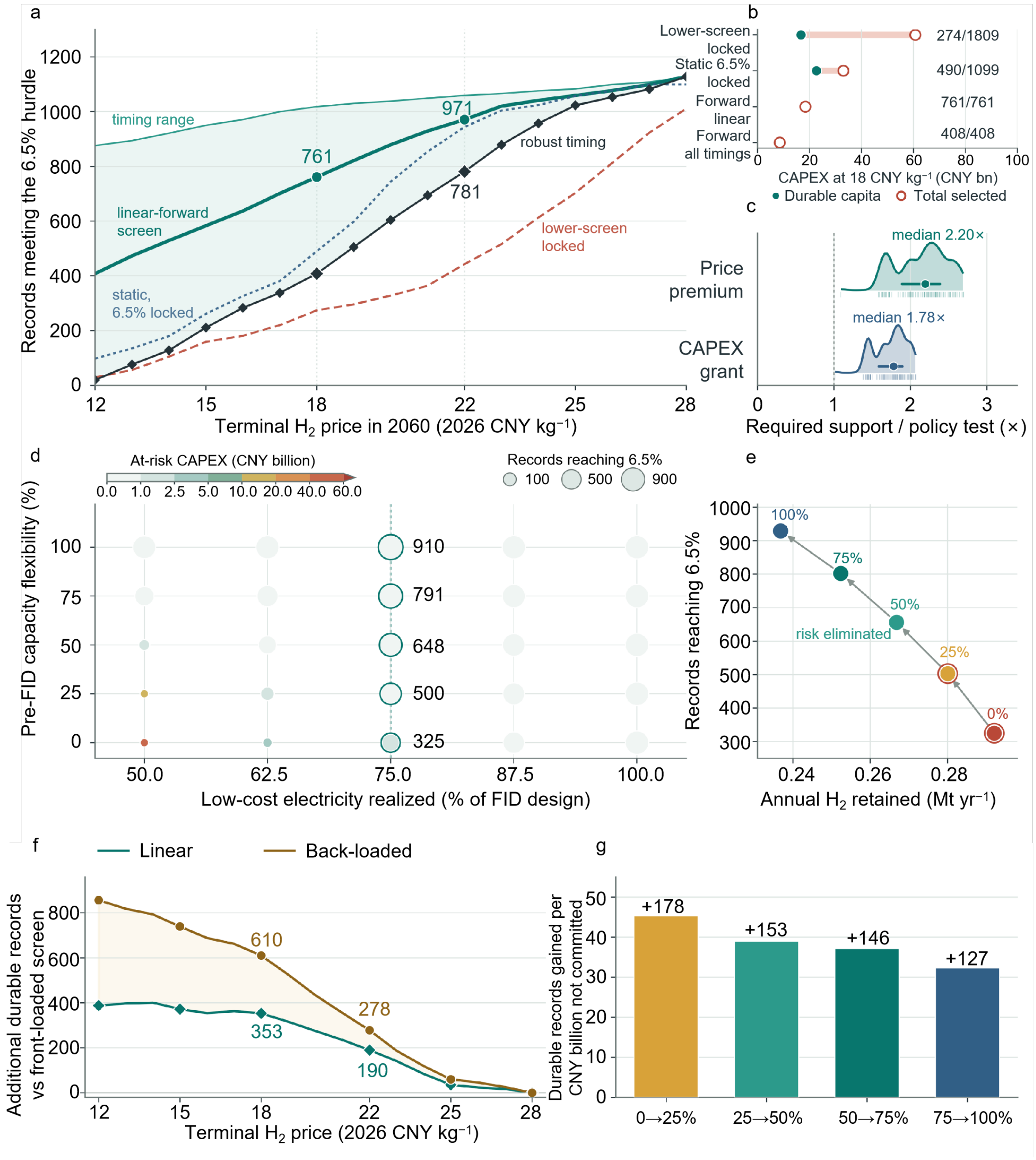

Figure 4: Price-informed screening and pre-investment capacity flexibility move exposure ahead of capital lock-in. **a**, Records remaining above 6.5% under alternative commitment rules across conditional terminal prices of 12–28 CNY kg^{-1}; the green band spans front-loaded to back-loaded timing, the heavy line denotes the linear path and black diamonds denote timing-robust screening. **b**, Durable and total selected capital under four commitment rules for the linear 18 CNY kg^{-1} path; right-hand labels give durable/selected records. **c**, Required 15-year price premiums and upfront capital grants normalised by illustrative policy tests; ridges show distributions, thick lines show interquartile ranges and circles mark medians. **d**, At-risk capital under the primary 1-MW minimum-build convention across resource realisation and pre-FID adjustability; colour gives at-risk capital, circle area gives records reaching 6.5%, and outlines and adjacent values identify the 75% resource slice. **e**, Continuous-downsizing trade-off at 75% resource realisation between annual hydrogen retained and records reaching 6.5%; labels give adjustability and coral outlines identify cases with residual exposure. **f**, Gain in durable records under linear and back-loaded timing relative to the worst-case front-loaded screen. Lines include every one-CNY endpoint from 12 to 28 CNY kg^{-1}, and markers identify the six labelled endpoints. **g**, Marginal durability yield of each 25-percentage-point increase in continuous pre-FID adjustability at 75% resource realisation. Each step releases the same 3.93 billion CNY of planned CAPEX from commitment. Bars report the additional records reaching 6.5% per CNY billion not committed, and labels above bars give the corresponding record gains; declining bar heights therefore indicate diminishing marginal durability gains per equal capital increment.